# Kaan Aytekin

+1 (585) 202-5147 • kaytekin@u.rochester.edu • Linkedin • www.kaanaytekin.com (Project Portfolio)

## EDUCATION

**University of Rochester** **Rochester, NY**
*Bachelor of Science, Mechanical Engineering* *Anticipated May 2026*

- GPA 3.94 out of 4.00, Dean's List all applicable semesters
- Davis UWC Scholar, Tau Beta Pi Scholar

**UWC Robert Bosch College** **Freiburg, Germany**
*IB Diploma* *May 2022*

## ENGINEERING EXPERIENCE (See Portfolio for more detail - www.kaanaytekin.com)

**ASML – Air Bearing Metrology and Testing Rig** **Rochester, NY**
*Team Member* *January 2026- Present*

- **Engineered a high-precision metrology rig** to characterize load capacity and sub-micron fly-height of commercial air bearings, enabling ASML to evaluate cost-effective off-the-shelf alternatives to custom components.
- **Conducted FEA analysis** (NASTRAN Sol 101, 103) on load application structure to ensure safety during peak loading conditions.

**UR Baja SAE (Offroad Car Collegiate Design Series)** **Rochester, New York**
*Chief Engineer and Drivetrain Project Team Co-Leader* *May 2024 – May 2025*

- **Directed a 30-member multi-disciplinary team** through complete vehicle design and fabrication, achieving a 40% increase in national competition rankings while reducing the project spending by 5%.
- **Designed the team's first-ever custom differential** along with a custom **transfer case and driveshaft** using Siemens NX. Used linear and multi step non-linear FEA to determine differential FoS.
- Validated gear-train integrity by applying AGMA standards and MATLAB to create solvers for bevel gear forces.

*Power Distribution Project Team Leader (Responsible of Axles)* *August 2023 – May 2024*

- Optimized axle and universal joint geometry using engineering analysis methods such as FEA (NASTRAN), dynamics simulations (Siemens Simcenter 3D), and solid mechanics principles, troubleshooting critical interference and loosening issues while achieving a 64% reduction in material costs.
- Conducted Multi-Body Physics sim. in Siemens NX to ensure the axles perform throughout the suspension RoM.
- Manufactured 50+ custom designed axle yokes in a 4-axis CNC mill, used a manual mill to manufacture axle flanges, inserts and more. Designed custom fixture and tooling for bending and CNC.

**Torsion Testing Rig for 4130 Steel Axles** **Rochester, NY**
*Author* *January 2024- May 2024*

- **Developed a custom torque testing rig** and testing methodology to measure the torsional yield characteristics of heat-treated 4130 steel (no-treatment, annealed and quenched samples) using strain gauges, NI DAQ devices, analyzing data using MATLAB and first principles engineering equations.
- Conclusively validated that targeted heat treatment statistically and significantly improves the torsional yield point of steel half-shaft connections, supporting component weight reduction in vehicle design.

**Laboratory for Laser Energetics (DoE funding $90M+, 400+ employees)** **Rochester, NY**
*Research Assistant* *March 2023 - Present*

- Conducted in-depth numerical calculations of free-free laser absorption rates in plasmas, incorporating quantum scattering theory and screening effects.
- Leveraged Python programming skills (NumPy, SciPy and Matplotlib), to perform numerical evaluations of Ordinary Differential Equations (ODEs) and integrals.
- Performed computations of 100 absorption correction tables (Gaunt Factor) using a Linux-based HPC cluster. Fitted a parametric analytical expression with a three-variable equation to extract data for integration into opacity and hydro-codes along with fusion experiments.
- Presented a poster in the student session of APS DPP 2024 and received the Outstanding Poster Award

## AWARDS

**APS DPP '24 Outstanding Poster Award** **Atlanta, GA**
*American Physical Society Division of Plasma Physics Annual Meeting* *October 2024*

- Awarded to top 5 student presenters out of approx. 150 presenters.

## SKILLS&INTERESTS

- **CAD:** Siemens NX, Solidworks, Fusion 360, PTC Creo. **Simulation/CAE:** Simcenter 3D, NASTRAN (linear, nonlinear, real eigenvalues), Multibody Dynamics, ANSYS Fluent (Workbench, Dynamic Meshing, VoF Mixtures) **Data Analysis/Coding:** MATLAB (Real Time Data Acquisition, NI, Simulink, Strain Gages), Python (NumPy, SciPy, Matplotlib, Large dataset analysis) **Other:** Opto-Mechanical Design (Portfolio), Test Engineering, Data Acquisition
- Turkish (Native), English (Fluent), German (B1)
- **Music,** played and recorded piano and guitar, led band as the club leader and the lead guitarist.